

August 6, 2026

Dear Hiring Team,

I am writing to apply for the Software Engineer - Canada position at DataVisor. With a Master of Science in Computer Science from the University of Calgary and experience building Python/C++ systems for machine learning, large-scale data processing, backend services, and performance-aware data infrastructure, I am excited by DataVisor's mission to use advanced AI systems for real-time fraud and risk detection.

My graduate research with BigGeo and Mitacs focused on scalable software for large Earth observation datasets. I built C++ and Python tools for processing, storing, and retrieving satellite imagery, developed encoding pipelines that convert raw imagery into structured tensor files, and improved processing performance through C++ multithreading. This work required careful attention to data quality, efficient storage, retrieval latency, and computational scalability, all of which connect closely to DataVisor's need for robust systems that process large data streams and support ML-driven decisions.

I also bring hands-on deep learning and model evaluation experience. In my DEM super-resolution project, I trained PyTorch ResNet models, built data collection and preprocessing workflows, and evaluated results with quantitative metrics designed for real-world reliability. As a Deep Learning Teaching Assistant with Neuromatch Academy, I guided students through PyTorch, CNNs, transformers, generative models, and end-to-end research projects. These experiences give me a strong foundation for contributing to ML platform work that must be technically sound, measurable, and useful in production-facing contexts.

In addition, I have professional backend development experience from Asr Gooyesh Pardaz, where I developed Django REST APIs backed by PostgreSQL, implemented authentication flows, integrated payment-service logic, and contributed to an AI-enabled chatbot system. While my production experience has been strongest in Python/C++ rather than Java, Kafka, Flink, or Spark, I have repeatedly ramped up on new technical stacks and enjoy working where software engineering, data systems, and machine learning meet.

Thank you for considering my application. I would welcome the opportunity to discuss how my background in scalable data processing, machine learning workflows, backend engineering, and performance-focused software can contribute to DataVisor's next-generation fraud detection platform.

Sincerely,
Amir Mirzai Golpayegani
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